242-243页8-11,16-18.
225-226页6,8; 244页21; 249页1,2.

242-8

Let $T \in \mathcal{L}(\mathbb{R}^3)$ represented by

$$\begin{pmatrix} 3 & -4 & -4 \\ -1 & 3 & 2 \\ 2 & -4 & -3 \end{pmatrix}.$$

Find non-zero vectors $\alpha_1, \cdots, \alpha_r$ satisfying the conditions of Theorem 3.
Proof: $f_T = (x-1)^3$, $p_T = (x-1)^2$. Hence $p_1 = (x-1)^2$ and $p_2 = x-1$. Let $\alpha_1 = (1,0,0)$, $\alpha_2 = (2,1,0)$, then $T\alpha_1 = (3,-1,2)$ so $R\alpha_1 = \langle \alpha_1, T\alpha_1 \rangle$ and $R\alpha_2 = \langle \alpha_2 \rangle$ are independent, forming a cyclic decomposition of T.

243-9

$$A = \begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ -3 & -3 & -5 \end{pmatrix}.$$

Find P such that $P^{-1}AP$ is in rational form.
Solution: $f_A = (x-1)(x+2)^2$, $p_A = (x-1)(x+2)$.

$$P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 3 \\ 0 & 0 & -3 \end{pmatrix}$$

243-10

Let $F \subset \mathbb{C}$ and $T \in \mathcal{L}(F^4)$ represented as

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & a & 2 & 0 \\ 0 & 0 & b & 2 \end{pmatrix}.$$

Find f_T. Consider the cases $a = b = 1$; $a = b = 0$; $a = 0, b = 1$. In each case find p_T and nonzero vectors forming a cyclic decomposition.
Solution: Clearly $f_T = (x-2)^4$.
Case1: $a = b = 1$, then $p_T = f_T = (x-2)^4$, and $\alpha_1 = (1,0,0,0)$ is cyclic.
Case2: $a = b = 0$, then $p_T = (x-2)^2$, and $\alpha_1 = e_1, \alpha_2 = e_3, \alpha_3 = e_4$ form a cyclic decomposition.
Case3: $a = 0, b = 1$. Then $p_T = (x-2)^2$, and $\alpha_1 = e_1, \alpha_2 = e_3$ form a cyclic decomposition.

243-11

Prove that if $A, B \in F^{3\times 3}$, then $A \sim B$ iff $p_A = p_B$ and $f_A = f_B$. Give an example which shows that this is false for 4×4 matrices.
Proof: $A \sim B \implies p_A = p_B, f_A = f_B$ is trivial.
Suppose $p_A = p_B, f_A = f_B$, then if $\deg p_A = 1$, A, B are similar to the same diagonal matrix, where the diagonal is the roots of f_A. If $\deg p_A = 2$, then A, B have the same rational form consisting of the companion matrices of p_A and f_A/p_A. If $\deg p_A = 3$, then they have the same rational form of a single companion matrix of p_A.
For 4×4 matrices, given $f_A = x^4$ and $p_A = x^2$, it can be factored as x^2, x, x or x^2, x^2. Hence

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

are not similar but $f_A = f_B$ and $p_A = p_B$.

243-16

Let $A \in \mathbb{R}^{n\times n}$ such that $A^2 + I = 0$. Prove that if $n = 2k$ then $A \sim B = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$ over $\mathbb{R}$.
Proof: $p_A = x^2 + 1 = p_B$. Since $\deg f_A = 2k$ and the only roots of f_A are $\pm i$, $f_A = (x^2+1)^k$, and the elementary divisors of A are just k multiples of p_A, the same with B. Hence $A \sim B$ over $\mathbb{R}$.

243-17

Let $T \in \mathcal{L}(V)$ with $\dim V < \infty$. Suppose that (a) p_T is a power of an irreducible polynomial; (b) $p_T = f_T$. Show that no non-trivial T-invariant subspace has a complementary $T-$invariant subspace.
Proof: Suppose $V = U \oplus W$ with U, W non-trivial $T-$invariant. Then $f_T = f_U f_W$ and $p_T = \operatorname{lcm}(p_U, p_W)$. Suppose $p_T = g^n$ with g irreducible, then $p_T = \operatorname{lcm}(p_U, p_W)$ implies (WLOG) $p_U = g^n$, so $f_T \neq p_T$, a contradiction.

243-18

If T is diagonalizable, then every $T-$invariant subspace has a complementary $T-$invariant subspace.
Proof: Let c_i be the eigenvalues of T, $E_i = \operatorname{Ker}(T - c_i I)$, then $V = E_1 \oplus \cdots \oplus E_n$. Recall for a $T-$invariant subspace W, $W = (W \cap E_1) \oplus \cdots \oplus (W \cap E_n)$. Clearly any subspace of E_i is T_i invariant, so we can take U_i as a complement of $W \cap E_i$ in E_i, then $U = U_1 \oplus \cdots \oplus U_n$ is a complementary $T-$invariant subspace.

226-6

Let $T \in \mathcal{L}(V)$ such that $rank(T) = 1$. Prove either T is diagonalizable or T is nilpotent, not both.
Proof: If T is diagonalizable and nilpotent, clearly $T = 0$ a contradiction.
Suppose $\operatorname{Im} T = \langle v \rangle$, then v is the only eigenvalue. If $Tv = 0$ then T is nilpotent. Otherwise suppose $Tv = cv$, then $x^2 - cx$ annihilates T, so $p_T \mid x(x-c)$. Hence T is diagonalizable.

226-8

Let $A \in F^{n\times n}$, $V = F^{n\times n}$, define $T \in \mathcal{L}(V)$ by $T(B) = AB - BA$. Prove that if A is nilpotent, then T is nilpotent.
Proof: $A \mapsto AB$ and $A \mapsto BA$ are both nilpotent, so T is also nilpotent, since $(x+y)^{2m} = \sum_k \binom{2m}{k} x^k y^{2m-k} = 0$ if $x^m = y^m = 0$.

244-21

Let $A \in \mathbb{R}^{n\times n}$, $T \in \mathcal{L}(\mathbb{R}^n)$ with $[T]_{\mathcal{B}} = A$, and $U \in \mathcal{L}(\mathbb{C}^n)$ with $[U]_{\mathcal{B}} = A$. Use the result of Ex243-20 to prove the following: If the only subspaces invariant under T are $\mathbb{R}^n$ and 0, then U is diagonalizable.
Proof: By Ex243-20 we know f_T is irreducible over $\mathbb{R}$. Hence f_T has only single roots in $\mathbb{C}$, so U is diagonalizable.

249-1

Let N_1, N_2 be 3×3 nilpotent matrices over F, prove that $N_1 \sim N_2$ iff they have the same minimal polynomial.
Proof: Since they are nilpotent, $f_1 = f_2 = x^3$. Hence by Ex243-11 $N_1 \sim N_2$ iff $p_1 = p_2$.

249-2

Use Ex249-1 and the Jordan form to prove the following: Let $A, B \in F^{n\times n}$ with $f_A = f_B = f = (x-c_1)^{d_1} \cdots (x-c_k)^{d_k}$, and $p_A = p_B$. If no $d_i > 3$ then A, B are similar.
Proof: Likewise the elementary divisors of A, B are uniquely determined.